

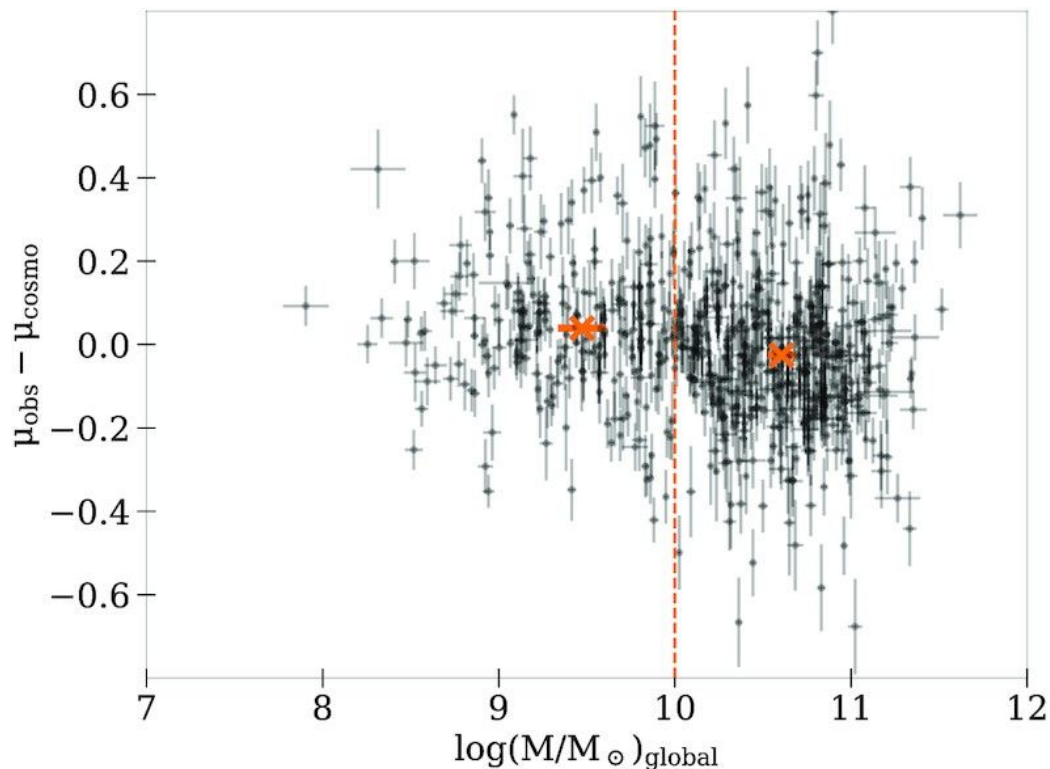
Environmental Dependence of the i-band SN Ia Secondary Maximum

Matt Grayling

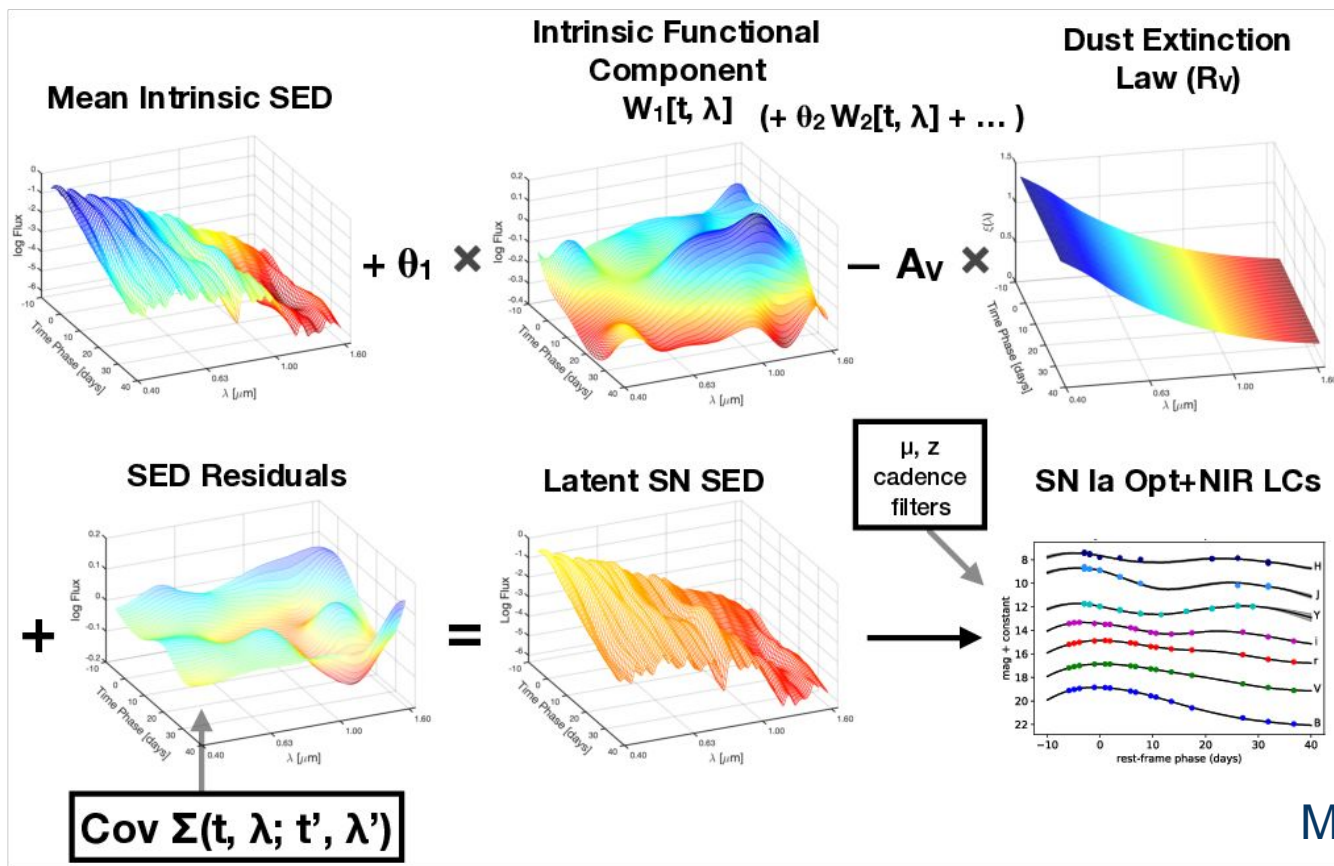


The 'Mass Step'

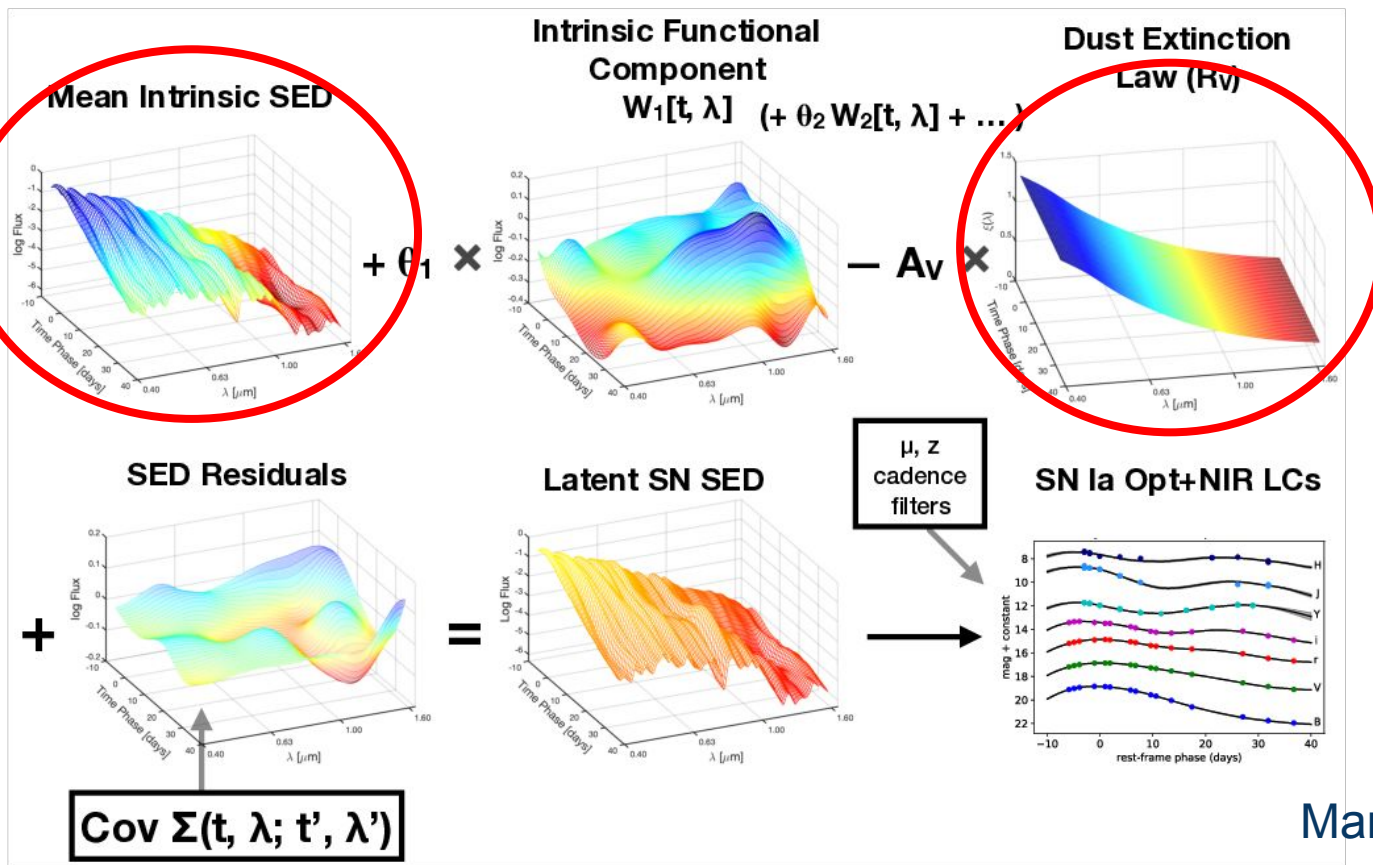
- Environmental dependence of *standardised* SN Ia peak absolute magnitude
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?



The BayeSN Model



Investigating the Mass Step

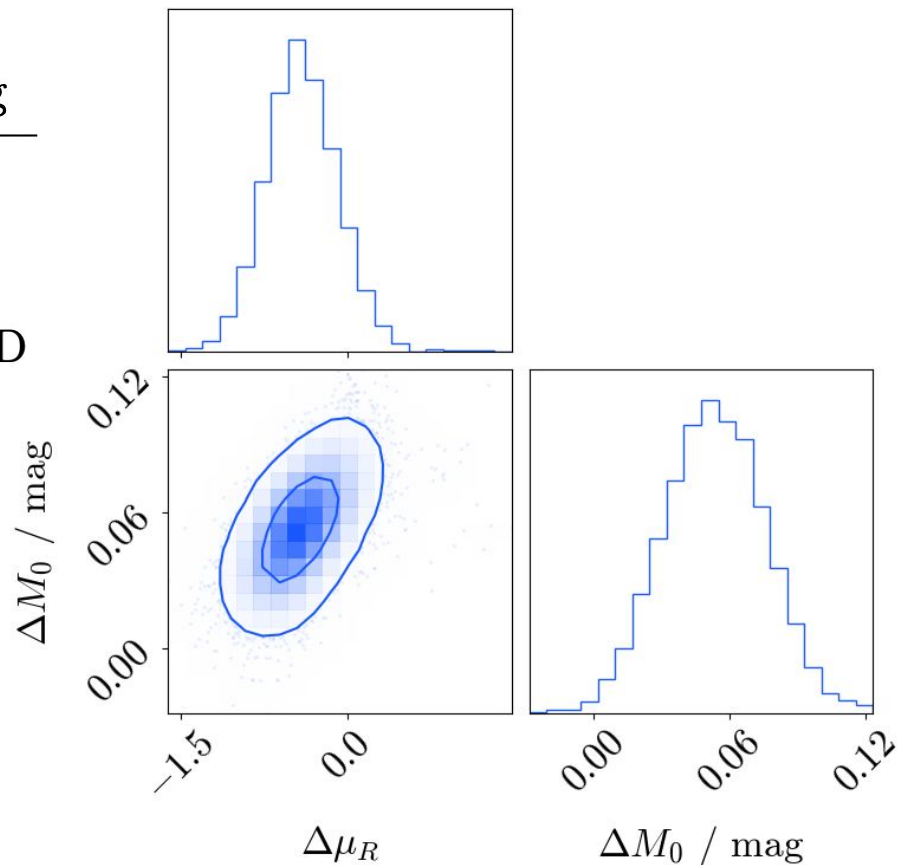


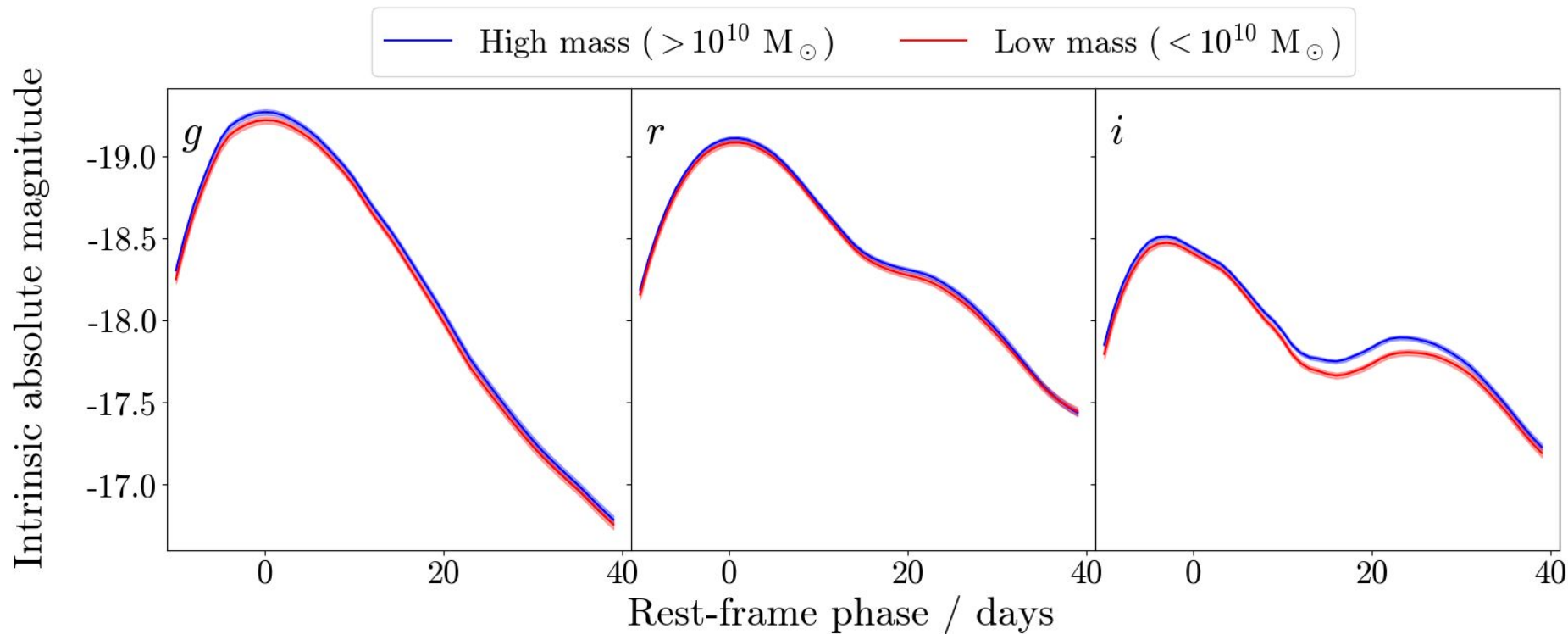
Dust or Intrinsic?

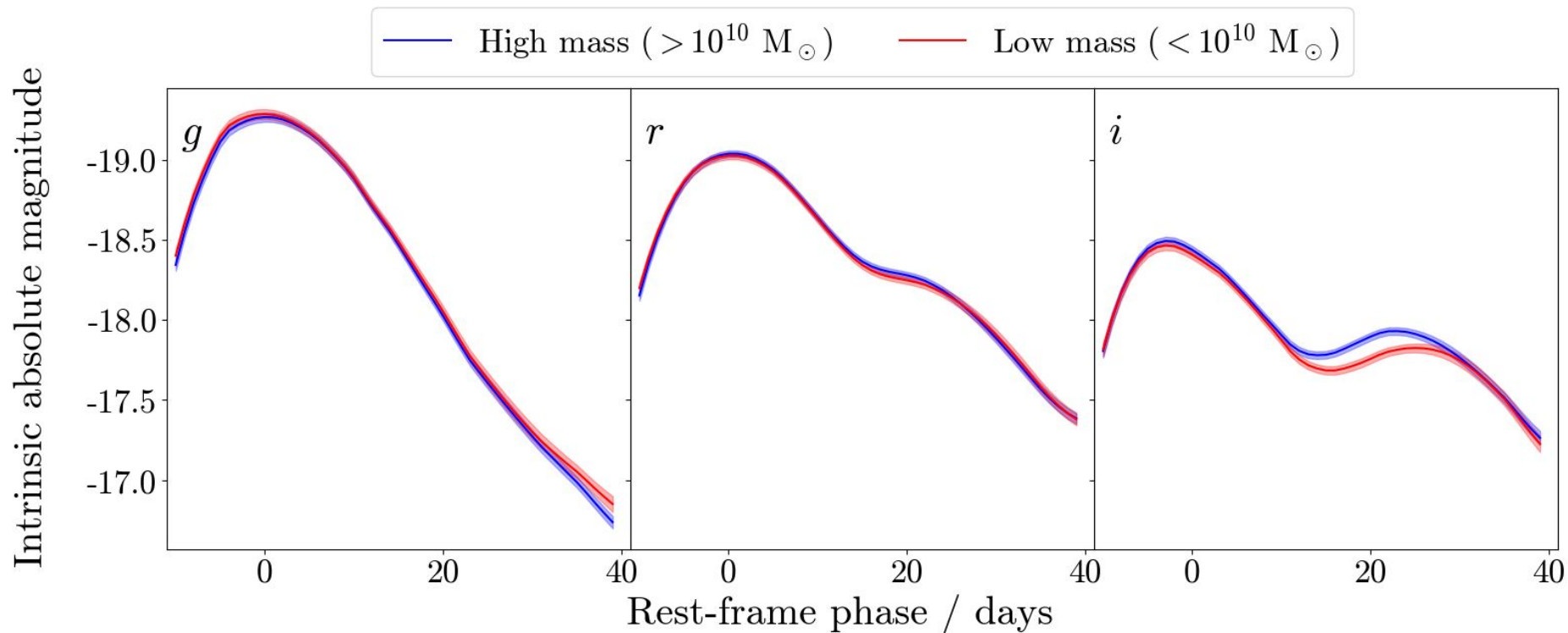
Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

*Combined = DES3YR+Foundation+PS1MD

Dust may play a role, but we do find evidence supporting an intrinsic contribution to the mass step



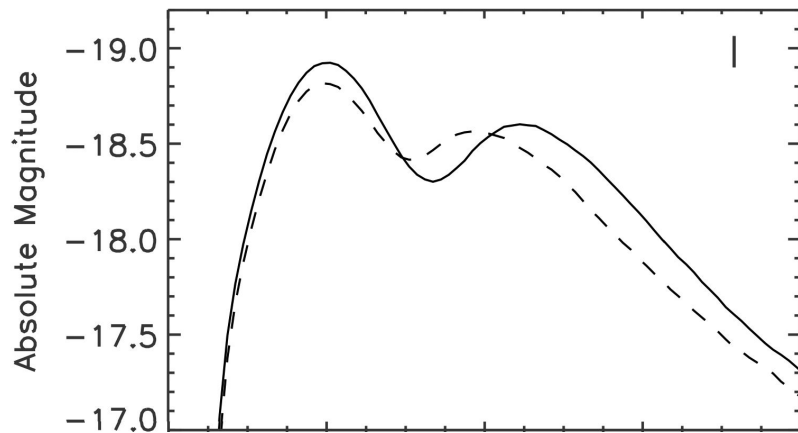




Grayling & Popovic (2024)

What's Causing the i-band Difference?

- Hopefully some people in this room might have some ideas!
- Kasen (2006) predicts that metallicity will have stronger impact on secondary maximum than around peak
- Difference specifically in i-band; Deckers et al. (2024) find differences between behaviour of second maximum in i and r-band
- Can't currently link this a specific spectral feature as BayeSN is only applied to photometry currently, but this will change in future!



From Kasen (2006); dashed line = higher metallicity

BayeSN Out Now!

- New GPU-accelerated, Python implementation of BayeSN out now! Use for population inference or individual light curve fitting
- github.com/bayesn/bayesn, pip installable, SNANA integration in progress
- New arxiv paper, validating BayeSN dust inference on SALT-based simulations and applying to DES5YR



MNRAS paper



New arxiv paper

Summary

- BayeSN out now, pip installable, can also be used for general light-curve fitting/distance estimation
- We find evidence to support an intrinsic component of the mass step though dust may also play a role
- We find a significant environmental-dependence of the intrinsic properties of SNe Ia, specifically on the i-band secondary maximum
- Lots of ongoing work for BayeSN e.g. training on spectra, SNANA integration, new dust laws



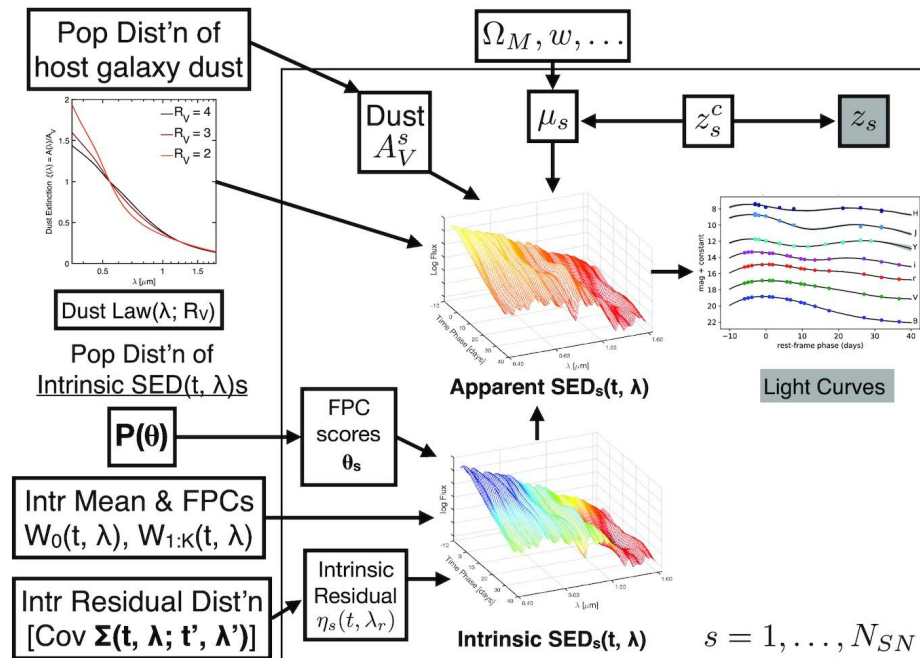
MNRAS paper



New arxiv paper

Advantages of BayeSN

- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation
- Full SED model
 - Uses full light curve in all bands rather than just single colour/magnitude pairs
- Fit for distance jointly with stretch/dust
 - No Tripp formula, distance marginalised over dust/intrinsic variation



Mandel+2020

Intrinsic Baseline Colour Curves

